

PROBRAMMING EXERCISES

- Problem 1** The following program is designed to find the area of a rectangle, the area of a circle, or the volume of a cylinder. However, (a) the statements are in the incorrect order; (b) the method calls are incorrect; (c) the logical expression in the while loop is incorrect; and (d) method definitions are incorrect. Rewrite the program so that it works correctly. Your program must be properly indented. (Note that the program is menu driven and allows the user to run the program as long as the user wishes.)

```
import java.util.*;

public class Ch7_PrExercise5
{
    static Scanner console = new Scanner(System.in);

    public static void main(String[] args)
    {
        double radius;
        double height;

        System.out.println("This program can calculate "
                           + "the area of a rectangle, the area "
                           + "of a circle, or volume of a cylinder.");
        System.out.println("To run the program enter: ");
        System.out.println("1: To find the area of rectangle.");
        System.out.println("2: To find the area of a circle.");
        System.out.println("3: To find the volume of a cylinder.");
        System.out.println("-1: To terminate the program.");
        choice = console.nextInt();
        System.out.println();

        int choice;
```

```

while (choice == -1)
{
    {
        case 1:
            System.out.print("Enter the radius of the base and "
                            + "the height of the cylinder: ");
            radius = console.nextDouble();
            height = console.nextDouble();
            System.out.println();

            System.out.printf("Area = %.2f%n",
                            circle(length, height));

            break;

        case 3:
            double length, width;
            System.out.print("Enter the radius of the circle: ");
            radius = console.nextDouble();
            System.out.println();

            System.out.printf("Area = %.2f%n", rectangle(radius));
            break;

        case 2:
            System.out.print("Enter the length and the width "
                            + "of the rectangle: ");
            length = console.nextDouble();
            width = console.nextDouble();
            System.out.println();

            System.out.printf("Volume = %.2f%n",
                            cylinder(radius, height));

            break;
        default:
            System.out.println("Invalid choice!");
    }
}
switch (choice)
{
}

```

```

        System.out.println("To run the program enter: ");
        System.out.println("2: To find the area of a circle.");
        System.out.println("1: To find the area of rectangle.");
        System.out.println("3: To find the volume of a cylinder.");
        System.out.println("-1: To terminate the program.");
        choice = console.nextInt();
        System.out.println();
    }

    public static double rectangle(double l, double w)
    {
        return l * w;
    }

    public static double circle(double r)
    {
        return Math.PI * r * r;
    }

    public static double cylinder(double bR, double h)
    {
        return Math.PI * bR * bR * h;
    }
}

```

Problem 2

The following formula gives the distance between two points (x_1, y_1) and (x_2, y_2) in the Cartesian plane:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Given the center and a point on a circle, you can use this formula to find the radius of the circle. Write a program that prompts the user to enter the center and a point on the circle. The program should then output the circle's radius, diameter, circumference, and area. Your program must have at least the following methods:

- a. **distance**: This method takes as its parameters four numbers that represent two points in the plane and returns the distance between them.
- b. **radius**: This method takes as its parameters four numbers that represent the center and a point on the circle, calls the method **distance** to find the radius of the circle, and returns the circle's radius.
- c. **circumference**: This method takes as its parameter a number that represents the radius of the circle and returns the circle's circumference. (If r is the radius, the circumference is $2\pi r$.)
- d. **area**: This method takes as its parameter a number that represents the radius of the circle and returns the circle's area. (If r is the radius, the area is πr^2 .)
- e. Assume that $\pi = 3.1416$.

Problem 3 A nonnegative integer is called a **palindrome** if it reads forward and backward in the same way. For example, the numbers 5, 121, 3443, and 123454321 are palindromes. Write a method that takes as input a nonnegative integer and returns **true** if the number is a palindrome; otherwise, it returns **false**. (When determining whether the number is a palindrome, do not convert the number into a string.) Also write a program to test your method.

Problem 4 Write a program that takes as input five numbers and outputs the mean (average) and standard deviation of the numbers. If the numbers are x_1, x_2, x_3, x_4 , and x_5 , then the mean is $x = (x_1 + x_2 + x_3 + x_4 + x_5) / 5$ and the standard deviation is:

$$s = \sqrt{\frac{(x_1 - x)^2 + (x_2 - x)^2 + (x_3 - x)^2 + (x_4 - x)^2 + (x_5 - x)^2}{5}}.$$

Your program must contain at least the following method: A method that calculates and returns the mean and a method that calculates the standard deviation.

Problem 5 The cost to become a member of a fitness center is as follows: (a) the Senior citizens discount is 30%; (b) if the membership is bought and paid for 12 or more months in advance, the discount is 15%; or (c) if more than 5 personal training sessions are purchased, the discount on each session is 20%.

Write a menu driven program that determines the cost of a new membership. Your program must contain a method that displays the general information about the fitness center and its charges, a method to get all the necessary information to determine the membership cost, and a method to determine the membership cost. Use appropriate parameters to pass information in and out of a method.

Problem 6 (**The box problem**) You have been given a flat cardboard of area, say, 70 square inches, to make an open box by cutting a square from each corner and folding the sides (see Figure 7-17). Your objective is to determine the dimension, that is, the length and width, and the side of the square to be cut from the corners so that the resulting box is of maximum volume.

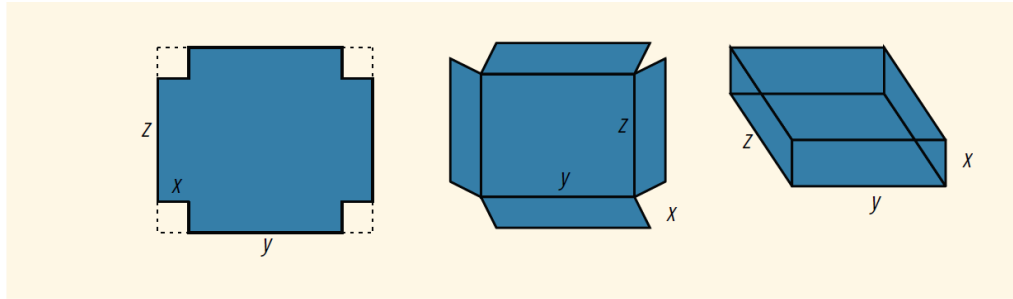


FIGURE 7-17 Cardboard box

Write a program that prompts the user to enter the area of the flat cardboard. The program then outputs the length and width of the cardboard and the length of the side of the square to be cut from the corner so that the resulting box is of maximum volume. Calculate your answer to three decimal places. Your program must contain a method that takes as input the length and width of the cardboard and returns the side of the square that should be cut to maximize the volume. The method also returns the maximum volume.

Problem 7

(The Power Station Problem) A power station is on one side of a river that is one-half miles wide, and a factory is 8 miles downstream on the other side of the river (see Figure 7-18). It costs \$7 per foot to run power lines overland and \$9 per foot to run them underwater. Your objective is to determine the most economical path to lay the power line. That is, determine how long the power line should run underwater and how long it should run over land, to achieve the minimum total cost of laying the power line.

Write a program that prompts the user to enter the following:

- The width of the river.
- The distance of the factory downstream on the other side of the river.
- Cost of laying the power line underwater.
- Cost of laying the power line overland.

The program then outputs the length of the power line that should run underwater and the length that should run over land, so the cost of constructing the power line is at the minimum. The program should also output the total cost of constructing the power line.

Problem 8

(Pipe Problem, requires trigonometry) A pipe is to be carried around the right-angled corner of two intersecting corridors. Suppose that the widths of the two intersecting corridors are 5 feet and 8 feet (see Figure 7-19). Your objective is to find the length of the longest pipe, rounded to the nearest foot, that can be carried level around the right-angled corner.

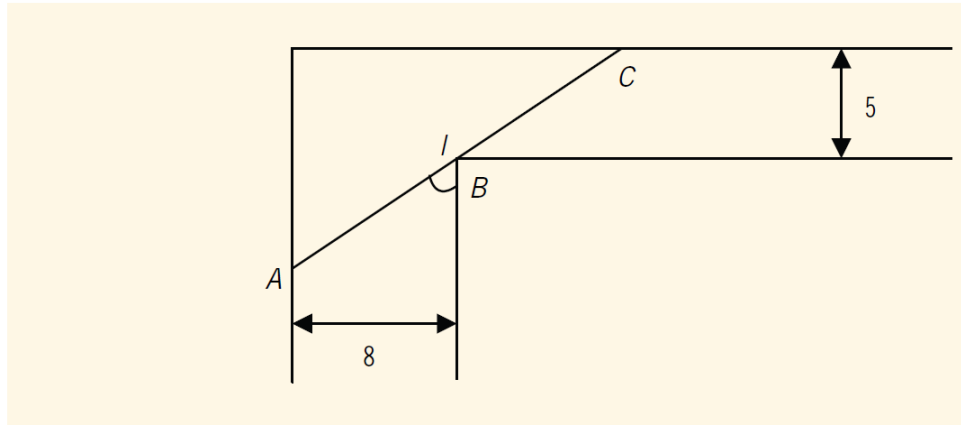


FIGURE 7-19 Pipe problem

Write a program that prompts the user to input the widths of both the hallways. The program then outputs the length of the longest pipe, rounded to the nearest foot, that can be carried level around the right-angled corner. (Note that the length of the pipe is given by $l = AB + BC = 8 / \sin \theta + 5 / \cos \theta$, where $0 < \theta < \pi/2$.)